

stpois: Fast Poisson event-history regression with high-dimensional fixed effects

Andreas Ljungström

andreas.ljungstrom@sofi.su.se

2026-07-13

Abstract Poisson regression on episode-split survival data is a workhorse of event-history analysis, but two practical obstacles limit its use with full-population register data: models with rich fixed-effect structures become slow or infeasible to estimate with explicit dummy variables, and continuous covariates prevent the collapsing of individual records to risk cells that otherwise makes such models fast. This article introduces the community-contributed command `stpois`, which fits Poisson event-history models directly on `stset` data through three estimation paths: an exact path that wraps `poisson` with automatic exposure; an exact high-dimensional fixed-effects path implemented via iteratively reweighted alternating projections in Mata, with OIM, robust, and cluster-robust standard errors that are numerically identical to the dummy-variable model; and two approximate fast paths that retain continuous covariates while estimating on collapsed cell data, either by folding first-stage continuous effects into a mutated exposure or by including within-cell moments as aggregation-bias corrections derived from a cumulant-generating-function expansion. I describe the methods, document their exactness properties and approximation error, and illustrate the command with examples.

1 Introduction

The equivalence between piecewise-constant hazard models and Poisson regression on episode-split data is one of the older and more useful results in event-history analysis (Holford 1980; Laird and Olivier 1981; Whitehead 1980). Splitting each subject's time at risk into episodes, treating the event indicator in each episode as a Poisson count, and entering the log of exposure time as an offset yields maximum-likelihood estimates of a proportional-hazards model with a hazard that is constant within episodes. In Stata, this model can be fitted with `poisson` after manually generating the exposure variable, or — for the single-episode exponential case — with `streg`, `distribution(exponential)`. The approach is standard in epidemiology (Clayton and Hills 1993; Carstensen 2007) and is covered at length by Rabe-Hesketh and Skrondal (2022).

Two practical obstacles arise when this framework meets modern administrative data. Full-population registers of the kind available in the Scandinavian countries routinely produce analysis files with millions or tens of millions of episode records, particularly after splitting on calendar time, age, and duration. First, models with rich fixed-effect structures — birth cohorts, municipalities, detailed occupational codes, or their interactions — become slow or infeasible when the fixed effects are entered as explicit dummy variables, since estimation time and memory grow with the square of the parameter count. The `ppmlhdfc` command (Correia et al. 2020) solves this problem for general Poisson models, building on the alternating-projections approach of Guimarães and Portugal (2010), but it is an external dependency

with its own installation chain, which matters in the locked-down secure environments where register data must typically be analyzed. Second, and less widely appreciated, the Poisson likelihood for purely categorical covariates depends on the data only through the sums of events and exposure within covariate cells. Estimation on the collapsed cell data is then *exact* and often several orders of magnitude faster. A single continuous covariate breaks this property, and analysts face an unattractive choice between discretizing the covariate and giving up the speed advantage entirely.

The command described in this article, `stpois`, addresses both obstacles within a single interface for `stset` data. It offers three estimation paths:

1. A **standard path** that wraps `poisson`, using the event indicator `_d` as outcome and time at risk `_t - _t0` as exposure. Coefficients and standard errors are identical to `streg, distribution(exponential)`.
2. An **exact high-dimensional fixed-effects path**, `absorb()`, implementing iteratively reweighted least squares with alternating-projections demeaning in Mata — the same mathematical approach as `pplhdfe`, but self-contained with no dependencies. OIM, robust, and cluster-robust standard errors are numerically identical to those from the corresponding dummy-variable model.
3. Two **approximate fast paths**, `fast(offset)` and `fast(moments)`, which retain continuous covariates while estimating on collapsed cell data. The first folds first-stage estimates of the continuous effects into a “mutated” exposure before collapsing; the second includes within-cell means, variances, and covariances of the continuous covariates as aggregation-bias corrections derived from a second-order cumulant-generating-function expansion.

The fast paths involve genuine approximations, and I am explicit throughout about what is exact and what is not. In particular, standard errors from both fast paths are conditional on first-stage estimates and do not propagate first-stage uncertainty; the moment-correction method additionally requires between-cell variation in the continuous covariates for the structural parameters to be well identified. Section 2 sets out the modeling framework, section 3 describes the collapsed-data methods, and section 4 describes the fixed-effects algorithm. Sections 5 and 6 document the syntax and illustrate the command, and section 7 reports validation results and timings.

2 Poisson event-history regression

Consider `stset` data in which record i contributes an event indicator $d_i \in \{0, 1\}$ (Stata’s `_d`) and a time at risk $t_i = _t - _t0$. Under a proportional-hazards model with hazard rate $\lambda_i = \exp(\mathbf{x}_i' \boldsymbol{\beta})$ constant within records, the log likelihood for right-censored data is, up to a constant,

$$\ell(\boldsymbol{\beta}) = \sum_i \{d_i \mathbf{x}_i' \boldsymbol{\beta} + d_i \log t_i - t_i \exp(\mathbf{x}_i' \boldsymbol{\beta})\},$$

which is proportional to the likelihood of a Poisson model for d_i with mean $\mu_i = t_i \exp(\mathbf{x}_i' \boldsymbol{\beta})$, that is, with $\log t_i$ as offset. Hence any software that fits Poisson regression with an offset fits this hazard model, and duration dependence or period effects are accommodated by episode splitting (`stsplit`) followed by indicator variables for the resulting time bands (Holford 1980; Laird and Olivier 1981). The Poisson “counts” here are 0/1 event indicators; nothing requires the outcome to be a genuine count, and the pseudo-maximum-likelihood results of Gourieroux et al. (1984) imply that the coefficient estimates are consistent whenever the log-linear mean is correctly specified, with robust standard errors available in the usual way (see also Santos Silva and Tenreiro 2006).

`stpois` fits exactly this model on the current `stset` data. On the standard path it constructs the exposure variable, calls `poisson`, and posts results with survival-flavored headers (numbers of subjects and failures,

time at risk). The point of the command, however, is not the convenience wrapper but the two departures from it, to which I now turn.

2.1 Exact collapsing with categorical covariates

When all covariates are categorical, the covariate vector takes a finite number of distinct values, indexing cells $j = 1, \dots, J$. The log likelihood above then depends on the data only through the within-cell sums $D_j = \sum_{i \in j} d_i$ and $T_j = \sum_{i \in j} t_i$:

$$\ell(\beta) = \sum_j \{D_j \mathbf{x}'_j \beta - T_j \exp(\mathbf{x}'_j \beta)\} + \text{const}.$$

(D_j, T_j) are sufficient statistics, and Poisson estimation on the J collapsed cells reproduces the individual-level maximum-likelihood estimates exactly — coefficients, standard errors, and all. With millions of episode records but a few hundred cells (period $\times$ education $\times$ region, say), the gain is dramatic, and this is the standard trick behind register-based rate modeling (Clayton and Hills 1993). A continuous covariate destroys the sufficiency because $\exp(\gamma x_i)$ no longer aggregates: $\sum_{i \in j} t_i \exp(\gamma x_i) \neq T_j \exp(\gamma \bar{x}_j)$ by Jensen's inequality. The two fast paths in `stpois` are two different responses to this problem.

3 Fast estimation with continuous covariates

Partition the covariates into continuous variables $\mathbf{X}_i$ with coefficients γ and categorical variables defining cells j with cell effects α_j , so that $\lambda_i = \exp(\gamma' \mathbf{X}_i + \alpha_{j(i)})$. In the `stpois` syntax, bare variable names are treated as continuous and `i.`-prefixed terms as categorical; the categorical variables define the collapse cells.

3.1 Two-stage offset: `fast(offset)`

The multiplicative structure of the log link, $\exp(\gamma' \mathbf{X}_i + \alpha_j) = \exp(\gamma' \mathbf{X}_i) \exp(\alpha_j)$, means that a *known* γ can be folded into the exposure. Stage 1 estimates $\hat{\gamma}$ by Poisson regression on the full individual-level data, including the categorical indicators to avoid omitted-variable bias. Stage 2 constructs the mutated exposure

$$\tilde{t}_i = t_i \exp(\hat{\gamma}' \mathbf{X}_i),$$

collapses d_i and $\tilde{t}_i$ to cell sums $(D_j, \tilde{T}_j)$, and fits the categorical-only Poisson model on the J cells with $\log \tilde{T}_j$ as offset.

Given $\hat{\gamma}$, this second stage is exact in the same sense as section 2.1: because the stage-1 model is the very model of interest, the stage-2 score equations for the cell parameters are the profile score equations of the full model, and the stage-2 coefficient estimates replicate the full maximum-likelihood estimates of the categorical parameters. In the validation exercise of section 7 they agree to six decimal places. What is *not* exact are the standard errors: the stage-2 variance matrix conditions on $\hat{\gamma}$ and therefore omits the first-stage sampling variation that a joint or two-step corrected estimator (Murphy and Topel 1985) would propagate. The constant's standard error is visibly too small in the examples below; the cell-contrast standard errors are much less affected because the first-stage error largely shifts all cells together. Since stage 1 runs on the full individual-level data, `fast(offset)` does not save time in a single estimation — its value lies in situations where the collapsed data are reused, as when many categorical specifications are explored over a fixed set of continuous controls, or when the collapsed file, unlike the microdata, may leave a secure server.

3.2 Moment correction: `fast(moments)`

The second method avoids the individual-level estimation entirely. The cell-aggregated mean under the model is

$$E[D_j] = \exp(\alpha_j) \sum_{i \in j} t_i \exp(\gamma' \mathbf{X}_i),$$

and the problematic term is the log of an average of exponentials — a cumulant generating function evaluated at γ . A second-order expansion around the within-cell distribution of $\mathbf{X}$ gives

$$\log \left\{ \frac{1}{T_j} \sum_{i \in j} t_i \exp(\gamma' \mathbf{X}_i) \right\} \approx \gamma' \boldsymbol{\mu}_j + \frac{1}{2} \gamma' \boldsymbol{\Sigma}_j \gamma,$$

where $\boldsymbol{\mu}_j$ and $\boldsymbol{\Sigma}_j$ are the (exposure-weighted, in principle; unweighted, as implemented) mean vector and covariance matrix of $\mathbf{X}$ within cell j . For two continuous covariates the correction terms are $\gamma_1 \mu_{1j} + \gamma_2 \mu_{2j} + \frac{1}{2} \gamma_1^2 \sigma_{1j}^2 + \frac{1}{2} \gamma_2^2 \sigma_{2j}^2 + \gamma_1 \gamma_2 \sigma_{12j}$. The expansion is exact when $\mathbf{X}$ is normal within cells and is otherwise a second-order approximation; an optional third-order term (option `skewness`) adds within-cell third central moments for skewed covariates.

`stpois`, `fast(moments)` computes the within-cell means, variances, and covariances in a single `collapse` pass and enters them as *free* regressors in the cell-level Poisson model, alongside the categorical terms. The coefficients on the means are the structural γ ; the coefficients on the variance and covariance terms are left unrestricted rather than constrained to $\gamma^2/2$ and $\gamma_1 \gamma_2$, which turns the constraint into a diagnostic: if the estimated coefficient on σ_{kj}^2 is close to $\hat{\gamma}_k^2/2$, the second-order approximation is internally consistent.

Two limitations deserve emphasis. First, as with `fast(offset)`, the standard errors condition on the moment estimates. Second — and this is the binding constraint in practice — the structural γ is identified only from *between-cell* variation in the moments. If the continuous covariates are (nearly) orthogonal to the cell structure, the cell means $\boldsymbol{\mu}_j$ barely vary, and $\hat{\gamma}$ is estimated imprecisely however large the underlying microdata. The simulation in section 7 illustrates this deliberately unfavorable case: with covariates assigned independently of cells, the categorical contrasts are recovered almost perfectly while the structural γ carries wide confidence intervals that nevertheless cover the truth. When the continuous covariates *do* vary across cells — the typical register-data situation, where age, income, and the like differ systematically across periods, regions, and education groups — identification is correspondingly stronger. Users should regard `fast(moments)` estimates of γ as by-products; the method's purpose is to protect the *categorical* estimates from aggregation bias while retaining collapsed-data speed, which in the timings below is roughly seven times the full maximum-likelihood path even at half a million observations, with the gap growing in the number of observations.

4 High-dimensional fixed effects: `absorb()`

The `absorb(varlist)` option fits the exact model with one or more sets of fixed effects without creating dummy variables, using iteratively reweighted least squares (IRLS) combined with weighted alternating-projections demeaning — the approach introduced to Stata by Guimarães and Portugal (2010) and refined by Correia et al. (2020). Each IRLS iteration:

1. forms the working variable $z_i = \eta_i + (d_i - \mu_i)/\mu_i$ and weights $w_i = \mu_i$, where η_i is the current linear predictor including the offset and $\mu_i = t_i \exp(\eta_i)$;
2. demeans z and each covariate within the groups of each absorbed variable in turn, iterating this Gauss–Seidel cycle to convergence (weighted alternating projections);
3. regresses the demeaned $\tilde{z}$ on the demeaned $\tilde{\mathbf{X}}$ by weighted least squares to update β ; and
4. recovers the fixed-effect contribution as the complement of the projection and updates η .

Convergence is declared when $\max_k |\Delta \beta_k| / (1 + |\beta_k|)$ falls below `tolerance()` (default 10^{-8}). The implementation is pure Mata with no dependencies.

Standard errors come from the partitioned information at convergence: $V = (\tilde{\mathbf{X}}'W\tilde{\mathbf{X}})^{-1}$ for `vce(oim)`, the HC1 sandwich with meat $\sum_i s_i s_i' \cdot n/(n-1)$ and scores $s_i = (d_i - \mu_i)\tilde{\mathbf{x}}_i$ for `vce(robust)`, and the $G/(G-1)$ -corrected clustered sandwich with scores summed within clusters for `vce(cluster clustvar)`. By the Frisch–Waugh–Lovell logic these are not approximations: all three match the corresponding dummy-variable `poisson` output at numerical precision, which the test suite verifies (section 7).

Groups of an absorbed variable in which no events occur are separated in the maximum-likelihood sense — their fixed effects diverge to $-\infty$ — and `stpois` drops such observations before estimation with a message, mirroring the practice of `ppmlhdfc` (Correia et al. 2020, who treat separation in far greater generality). The inner demeaning loop costs $O(n)$ per absorbed variable and pass, which is efficient for the moderate group counts typical of event-history applications (periods, cohorts, regions, occupations). For extreme cases — say, person fixed effects with millions of levels — `ppmlhdfc`’s accelerated solver remains the better tool, and `stpois` makes no attempt to compete there.

5 The `stpois` command

5.1 Syntax

```
stpois varlist [if] [in] [weight] [, irr nolog level(#) noconstant
      vce(vcetype) robust cluster(varname)
      absorb(varlist) tolerance(#) maxiter(#)
      fast(offset | moments) skewness poisson_options ]
```

The data must be `stset`; see [ST] `stset`. `fweights`, `iweights`, and `pweights` are allowed. `fast()` and `absorb()` may not be combined. In the fast paths, bare variable names are treated as continuous and `i.`-prefixed variables as categorical grouping variables; interactions are not supported there.

5.2 Options

irr reports exponentiated coefficients (incidence-rate ratios, equal to hazard ratios in this model). Display only.

nolog suppresses iteration logs, including the progress notes of the fast and HDFE paths.

level(#) sets the confidence level; the default is `level(95)`.

noconstant suppresses the constant term.

vce(vcetype), **robust**, and **cluster(varname)** specify the variance estimator. On the standard path these are passed through to `poisson` and accept anything `poisson` accepts. With `absorb()`, `vce(oim)` (the default), `vce(robust)`, and `vce(cluster clustvar)` are supported, all numerically exact relative to the dummy-variable model.

absorb(varlist) absorbs the fixed effects of one or more categorical variables via IRLS with alternating-projections demeaning; see section 4.

tolerance(#) and **maxiter(#)** control the IRLS convergence criterion and iteration cap for `absorb()`; defaults are `1e-8` and `100`.

fast(offset | moments) selects a collapsed-data fast path; see section 3. **skewness** adds third-order moment corrections under `fast(moments)`.

Any remaining options on the standard path are passed to `poisson`.

5.3 Stored results

`stpois` stores the usual `e(b)`, `e(V)`, `e(sample)`, log likelihood, and model-test scalars, plus survival-specific scalars `e(N_sub)` (subjects), `e(N_fail)` (failures), and `e(risk)` (total time at risk). The fast paths add `e(N_cells)` and the macros `e(cont_vars)` and `e(cat_vars)`; `absorb()` adds `e(absorb)` and, with clustering, `e(N_clust)` and `e(clustvar)`. `e(predict)` is set to `stpois_p`, so `predict` offers the linear predictor (`xb`, the default, including the log-exposure offset unless `nooffset`), predicted numbers of events (`n`), hazard rates (`hazard` or `hr`), and survival probabilities (`survival`). Standard postestimation commands (`margins`, `lincom`, `test`, `estimates`) work as usual, with the caveat that likelihood-based comparisons across estimation paths are not meaningful because the collapsed-data likelihoods differ from the individual-level likelihood by path-specific constants.

6 Examples

6.1 Standard path

The default path reproduces `streg`, `distribution(exponential)` exactly (the log likelihoods differ by a constant because `streg` includes the $d_i \log t_i$ terms):

```
. webuse stan3
. stset t1, failure(died) id(id)
. stpois age posttran, nolog
```

Poisson event-history regression

No. of subjects =	103	Number of obs =	172
No. of failures =	75		
Time at risk =	31938	LR chi2(2) =	36.75
Log likelihood =	-233.83404	Prob > chi2 =	0.0000

	_d	Coefficient	Std. err.	z	P> z	[95% conf. interval]
age		.0605817	.015235	3.98	0.000	.0307216 .0904417
posttran		-1.285206	.2377048	-5.41	0.000	-1.751099 -.8193135
_cons		-7.8376	.7094787	-11.05	0.000	-9.228153 -6.447047

6.2 Absorbing fixed effects

`absorb()` returns the same coefficients and standard errors as the explicit dummy-variable model, without estimating the absorbed effects:

```
. stpois age posttran, absorb(surgery) nolog
```

Poisson EHA – HDFE (absorb: surgery)

No. of subjects =	103	Number of obs =	172
No. of failures =	75		
Time at risk =	31938	LR chi2(2) =	49.18
Log likelihood =	-227.61949	Prob > chi2 =	0.0000

	_d	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	age	.0610952	.014454	4.23	0.000	.0327659	.0894244
	posttran	-1.141144	.2402034	-4.75	0.000	-1.611934	-.6703538

For comparison, `poisson _d age posttran i.surgery, exposure(exp)` on the same sample gives `age` = .0610952 (SE .014454) and `posttran` = -1.141144 (SE .2402034): identical to all printed digits.

6.3 Fast paths on a large synthetic dataset

The following simulation generates 500,000 subjects in 200 risk cells (4 education × 5 region × 10 period), with two continuous covariates (`age`, `income`) that affect the hazard but are assigned independently of the cells:

```
. clear
. set obs 500000
. set seed 20260713
. gen edu = ceil(4 * runiform())
. gen region = ceil(5 * runiform())
. gen period = ceil(10 * runiform())
. gen age = 30 + 20 * runiform()
. gen income = rnormal(10, 2)
. gen lambda = exp(-6 + 0.03*age + 0.05*income
> + 0.3*(edu==2) + 0.5*(edu==3) - 0.2*(edu==4)
> + 0.1*(region==2) - 0.2*(region==3) + 0.2*(period >= 6))
. gen ptime = 0.5 + 0.5 * runiform()
. gen failed = (runiform() < 1 - exp(-lambda * ptime))
. gen id2 = _n
. stset ptime, failure(failed) id(id2)
```

This yields 6,759 events. Estimating the full model and both fast paths:

```
. stpois age income i.edu i.region i.period, nolog
. stpois age income i.edu i.region i.period, fast(offset) nolog
. stpois age income i.edu i.region i.period, fast(moments) nolog
```

The table below extracts selected coefficients (standard errors in parentheses):

	Full MLE	fast(offset)	fast(moments)
2.edu	0.3071 (0.0353)	0.3071 (0.0353)	0.3049 (0.0356)
3.edu	0.5094 (0.0339)	0.5094 (0.0339)	0.5059 (0.0342)
3.region	-0.1654 (0.0401)	-0.1654 (0.0401)	-0.1636 (0.0405)
9.period	0.2175 (0.0538)	0.2175 (0.0538)	0.2266 (0.0557)
age (γ_1)	0.0265 (0.0021)	— (stage 1)	0.0909 (0.1091)
income (γ_2)	0.0518 (0.0061)	— (stage 1)	-0.1341 (0.3322)
Estimation time	2.21 s	1.94 s	0.33 s

Three features of this table repay attention. First, `fast(offset)` reproduces the full maximum-likelihood categorical coefficients to six decimal places, as the sufficiency argument of section 3.1 predicts, and its cell-contrast standard errors are correct to the printed digits; only the constant’s standard error (not shown) is understated. Second, `fast(moments)` protects the categorical contrasts — all within about half a standard error of the full-MLE values — even though its structural γ estimates are, by design of this simulation, weakly identified: the covariates were assigned independently of the cells, so the cell means carry almost no information, and the wide confidence intervals (which cover the true 0.03 and 0.05) signal this honestly. Third, the timing gains at this scale are modest for `fast(offset)` (whose stage 1 runs on the microdata) and a factor of about seven for `fast(moments)`; both grow with the number of observations, since the cell-level estimation cost is fixed.

6.4 Episode-split register-style workflow

In applications, the typical sequence splits spells on calendar time or age and then estimates piecewise-constant hazard models:

```
. stsplit period, at(1990(5)2020) after(time=entry)
. stpois i.period age income i.edu, irr
. stpois age income, absorb(period edu region) vce(cluster region)
. stpois age income i.period i.edu i.region, fast(moments)
```

7 Validation and performance

The package ships with a test suite (`tests/test_stpois.do`) that runs on the `stan3` heart-transplant data and on simulated data. It verifies that (i) the standard path converges and matches `streg, distribution(exponential)`; (ii) `absorb()` matches the explicit dummy-variable `poisson` model on coefficients *and* standard errors to within 10^{-4} (the observed agreement is on the order of 10^{-8}), for OIM and robust variance estimators, with the cluster estimator covered by the same sandwich code path; (iii) `fast(offset)` reproduces full-MLE categorical coefficients on collapsed cells; and (iv) `fast(moments)` produces the structural-coefficient machinery and recovers signs on simulated data. All tests pass on Stata 18/MP; the command requires version 14.

The timings in section 6.3 were obtained on a single run (Stata MP, Apple silicon) and should be read as indicative. The general pattern is robust, however: collapsed-cell estimation time is essentially independent of the microdata size, so the advantage of `fast(moments)` — and of `fast(offset)` when the collapse is reused — grows linearly in the number of episode records. For register applications with 10^7 or more episodes and a few hundred cells, cell-level estimation turns minutes into fractions of a second, which changes what is practical to do interactively: specification searches over the categorical structure, bootstrap loops over the collapsed data, or estimation inside secure environments with tight session limits.

8 Conclusion

`stpois` packages three things that belong together but have hitherto required three different workflows: the Poisson formulation of event-history models on `stset` data, exact absorption of high-dimensional fixed effects without external dependencies, and collapsed-data estimation that retains continuous covariates. The exact paths are exact — the test suite holds the `absorb()` variance estimators to the dummy-variable benchmark, not merely the coefficients — and the approximate paths are explicit about their approximations: standard errors conditional on first-stage estimates, and, for `fast(moments)`, structural parameters identified only from between-cell moment variation. The natural extensions are a Murphy–Topel correction for the two-step standard errors and exposure-weighted within-cell moments,

both of which fit the existing architecture. The command, its help file, and the test suite are available from the author's repository and are installable with `net install`.

Acknowledgments

I thank colleagues at SOFI for discussions on collapsed-data estimation with Swedish register data.

References

- Carstensen, Bendix. 2007. "Age-Period-Cohort Models for the Lexis Diagram." *Statistics in Medicine* 26 (15): 3018–45.
- Clayton, David, and Michael Hills. 1993. *Statistical Models in Epidemiology*. Oxford University Press.
- Correia, Sergio, Paulo Guimarães, and Tom Zylkin. 2020. "Fast Poisson Estimation with High-Dimensional Fixed Effects." *Stata Journal* 20 (1): 95–115.
- Gourieroux, Christian, Alain Monfort, and Alain Trognon. 1984. "Pseudo Maximum Likelihood Methods: Applications to Poisson Models." *Econometrica* 52 (3): 701–20.
- Guimarães, Paulo, and Pedro Portugal. 2010. "A Simple Feasible Procedure to Fit Models with High-Dimensional Fixed Effects." *Stata Journal* 10 (4): 628–49.
- Holford, Theodore R. 1980. "The Analysis of Rates and of Survivorship Using Log-Linear Models." *Biometrics* 36 (2): 299–305.
- Laird, Nan, and Donald Olivier. 1981. "Covariance Analysis of Censored Survival Data Using Log-Linear Analysis Techniques." *Journal of the American Statistical Association* 76 (374): 231–40.
- Murphy, Kevin M., and Robert H. Topel. 1985. "Estimation and Inference in Two-Step Econometric Models." *Journal of Business & Economic Statistics* 3 (4): 88–97.
- Rabe-Hesketh, Sophia, and Anders Skrondal. 2022. *Multilevel and Longitudinal Modeling Using Stata*. 4th ed. Stata Press.
- Santos Silva, J. M. C., and Silvana Tenreyro. 2006. "The Log of Gravity." *Review of Economics and Statistics* 88 (4): 641–58.
- Whitehead, John. 1980. "Fitting Cox's Regression Model to Survival Data Using GLIM." *Journal of the Royal Statistical Society. Series C (Applied Statistics)* 29 (3): 268–75.

About the author

Andreas Ljungström is a researcher at the Swedish Institute for Social Research (SOFI), Stockholm University. His work concerns social stratification and intergenerational mobility, using full-population Swedish register data.